

Overview of Mathematical optimization

Subject: Mathematical optimization

1.

argmax_{x,y} x cos y, subject to: $x \in [-5, 5], y \in \mathbb{R}$

2.

Civil engineering Optimization has been widely used in civil engineering. Construction management and transportation engineering are among the main branches of civil engineering that heavily rely on optimization. The most common civil engineering problems that are solved by optimization are cut and fill of roads, life-cycle analysis of structures and infrastructures, resource leveling, water resource allocation, traffic management and schedule optimization.

3.

Given: a function $f : A \rightarrow \mathbb{R}$ from some set A to the real numbers
Sought: an element $x_0 \in A$ such that $f(x_0) \leq f(x)$ for all $x \in A$ ("minimization") or such that $f(x_0) \geq f(x)$ for all $x \in A$ ("maximization"). Such a formulation is called an optimization problem or a mathematical programming problem (a term not directly related to computer programming, but still in use for example in linear programming – see History below). Many real-world and theoretical problems may be modeled in this general framework. Since the following is valid:

4.

it suffices to solve only minimization problems. However, the opposite perspective of considering only maximization problems would be valid, too. Problems formulated using this technique in the fields of physics may refer to the technique as energy minimization, speaking of the value of the function f as representing the energy of the system being modeled. In machine learning, it is always necessary to continuously evaluate the quality of a data model by using a cost function where a minimum implies a set of possibly optimal parameters with an optimal (lowest) error. Typically, A is some subset of the Euclidean space $\mathbb{R}^n$, often specified by a set of constraints, equalities or inequalities that the members of A have to satisfy. The domain A of f is called the search space or the choice set, while the elements of A are called candidate solutions or feasible solutions. The function f is variously called an objective function, criterion function, loss function, cost function (minimization), utility function or fitness function (maximization), or, in certain fields, an energy function or energy functional. A feasible solution that minimizes (or maximizes) the objective function is called an optimal solution. In mathematics, conventional optimization problems are usually stated in terms of minimization. A local minimum x^* is defined as an element for which there exists some $\delta > 0$ such that

5.

Operations research Another field that uses optimization techniques extensively is operations research. Operations research also uses stochastic modeling and simulation to support improved decision-making. Increasingly, operations research uses stochastic programming to model dynamic

decisions that adapt to events; such problems can be solved with large-scale optimization and stochastic optimization methods.

6.

Mathematical optimization (alternatively spelled optimisation) or mathematical programming is the selection of a best element, with regard to some criteria, from some set of available alternatives. It is generally divided into two subfields: discrete optimization and continuous optimization. Optimization problems arise in all quantitative disciplines from computer science and engineering to operations research and economics, and the development of solution methods has been of interest in mathematics for centuries. In the more general approach, an optimization problem consists of maximizing or minimizing a real function by systematically choosing input values from within an allowed set and computing the value of the function. The generalization of optimization theory and techniques to other formulations constitutes a large area of applied mathematics.

7.

This denotes the minimum value of the objective function $x^2 + 1$, when choosing x from the set of real numbers $\mathbb{R}$. The minimum value in this case is 1, occurring at $x = 0$. Similarly, the notation

8.

Major subfields Convex programming studies the case when the objective function is convex (minimization) or concave (maximization) and the constraint set is convex. This can be viewed as a particular case of nonlinear programming or as generalization of linear or convex quadratic programming. Linear programming (LP), a type of convex programming, studies the case in which the objective function f is linear and the constraints are specified using only linear equalities and inequalities. Such a constraint set is called a polyhedron or a polytope if it is bounded. Second-order cone programming (SOCP) is a convex program, and includes certain types of quadratic programs. Semidefinite programming (SDP) is a subfield of convex optimization where the underlying variables are semidefinite matrices. It is a generalization of linear and convex quadratic programming. Conic programming is a general form of convex programming. LP, SOCP and SDP can all be viewed as conic programs with the appropriate type of cone. Geometric programming is a technique whereby objective and inequality constraints expressed as posynomials and equality constraints as monomials can be transformed into a convex program. Integer programming studies linear programs in which some or all variables are constrained to take on integer values. This is not convex, and in general much more difficult than regular linear programming. Quadratic programming allows the objective function to have quadratic terms, while the feasible set must be specified with linear equalities and inequalities. For specific forms of the quadratic term, this is a type of convex programming. Fractional programming studies optimization of ratios of two nonlinear functions. The special class of concave fractional programs can be transformed to a convex optimization problem. Nonlinear programming studies the general case in which the objective function or the constraints or both contain nonlinear parts. This may or may not be a convex program. In general, whether the program is convex affects the difficulty of solving it. Stochastic programming studies the case in which some of the constraints or parameters depend on random variables. Robust optimization is, like stochastic programming, an attempt to capture uncertainty in the data underlying the optimization problem. Robust optimization aims to find solutions that are valid under all possible realizations of the uncertainties defined by an

uncertainty set. Combinatorial optimization is concerned with problems where the set of feasible solutions is discrete or can be reduced to a discrete one. Stochastic optimization is used with random (noisy) function measurements or random inputs in the search process. Infinite-dimensional optimization studies the case when the set of feasible solutions is a subset of an infinite-dimensional space, such as a space of functions. Heuristics and metaheuristics make few or no assumptions about the problem being optimized. Usually, heuristics do not guarantee that any optimal solution need be found. On the other hand, heuristics are used to find approximate solutions for many complicated optimization problems. Constraint satisfaction studies the case in which the objective function f is constant (this is used in artificial intelligence, particularly in automated reasoning). Constraint programming is a programming paradigm wherein relations between variables are stated in the form of constraints. Disjunctive programming is used where at least one constraint must be satisfied but not all. It is of particular use in scheduling. Space mapping is a concept for modeling and optimization of an engineering system to high-fidelity (fine) model accuracy exploiting a suitable physically meaningful coarse or surrogate model. In a number of subfields, the techniques are designed primarily for optimization in dynamic contexts (that is, decision making over time):